

FIRST CODES:-

```
int rec(int n)
{
    if (n < 1)
        return 1;
    return n + rec(n-1) + rec(n-2);
}
```

SECOND CODES:-

```
int rec-iterative(int n)
{
    int i, f_n = 1, f_n1 = 1, f_n2 = 1;
    for (i = 1, i <= n, i++)
    {
        f_n = i + f_n1 + f_n2;
        f_n2 = f_n1;
        f_n1 = f_n;
    }
}
```

$$n=0 \Rightarrow \text{result} = 1$$

$$n=1 \Rightarrow 1 + \text{rec}(1-1) + \text{rec}(1-2) = 3$$

$$n=2 \Rightarrow 2 + \text{rec}(2-1) + \text{rec}(2-2) = 6$$

$$n=3 \Rightarrow 3 + \text{rec}(3-1) + \text{rec}(3-2) = 12$$

Example =

```
int F_filter (unsigned int n)
```

```
{ int k=0;
```

```
while (n>0){
```

```
    k+=n%3;
```

```
    n/=3;
```

```
}
```

```
return k;
```

```
int F (unsigned int n)
```

```
{ if (n<3)
```

```
    return n;
```

```
    return n%3+F(n/3);
```

```
}
```

verilen sayının 3' lük tabanındaki rakamları toplamı

OUTPUT

$n=0 \Rightarrow k=0$

1

$n=1 \Rightarrow k=1$

2

$n=2 \Rightarrow k=2$

2

$n=3 \Rightarrow k=0, n=1 \Rightarrow k=0+1=1$

1

$n=4 \Rightarrow k=1, n=1 \Rightarrow k=1+1=2$

2

$n=5 \Rightarrow k=2, n=1 \Rightarrow k=2+1=3$

3

$n=6 \Rightarrow k=0, n=2 \Rightarrow k=0+2=2$

2

$n=7 \Rightarrow k=1, n=2 \Rightarrow k=1+2=3$

3

$n=8 \Rightarrow k=2, n=2 \Rightarrow k=2+2=4$

4

$n=9 \Rightarrow k=0, n=3 \Rightarrow k=0+3=3$

3


```

int Rec_LCM(int m, int n, int common)
{
    if (m == 1 && n == 1)
        return 1;
    if (n % common == 0 || m % common == 0)
    {
        if (n % common == 0)
            n = n / common;
        if (m % common == 0)
            m = m / common;
        return Rec_LCM(m, n, common);
    }
    else
        return Rec_LCM(m, n, common + 1);
}

int LCM(int m, int n)
{
    return Rec_LCM(m, n, 2);
}

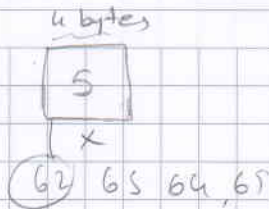
main()
{
    printf("%d", LCM(10, 15));
}

```

LCM(10, 15) $\xrightarrow{(5)}$ Rec_LCM(10, 15, 2) $\xrightarrow{(2)}$ Rec_LCM(5, 15, 2) $\xrightarrow{(4)}$ Rec_LCM(5, 3, 4) $\xleftarrow{(1)}$ Rec_LCM(5, 5, 3) $\xleftarrow{(3)}$ Rec_LCM(5, 15, 3) $\xrightarrow{(1)}$ Rec_LCM(5, 5, 5) $\xrightarrow{(2)}$ Rec_LCM(1, 1, 5) $\xrightarrow{(1)}$ 1 $\xrightarrow{(6)}$ print "1"

POINTERS

```
int x=5;
char y='A'
```



```
printf("%d %d", x, &x);
```

⇒ screen = 5 62

```
printf("%c %d", y, &y);
```

⇒ screen = A 100

```
printf("Adress of x is %p", &x);
```

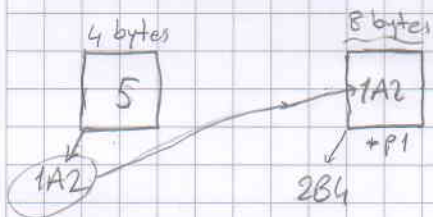
%p → hexadecimal notation

Definition of pointers:

```
int x; // x is a variable
```

```
int *pt; // pt is a pointer of integer
```

```
pt = &x;
```



x 'in' degeri 5, adresi 1A2
pt 'in' " 1A2, " 2B4

```
printf("%p\n", &x);
```

→ 1A2

```
printf("%p\n", pt);
```

→ 1A2

```
printf("%p\n", &pt);
```

→ 2B4

```
char ch; // ch is a character
```

```
char *p; // p is a pointer of character
```

```
int x=5
```

```
int *pt=&x;
```

*p

```
printf("%d", *pt);
```

→ 5

```
*pt=10;
```

(pt adresinin içindeki değeri 10 yap demek)

```
printf("%d: x) ;
```

→ 10